

Computer-aided simulations - Third Delivery

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Abstract—This document describes the progress of laboratory activities from 10 to 12 of the *Computer-aided simulations lab* course. In particular, these laboratories cover 3 different topics, respectively: Galton-Watson processes, epidemic compartmental models, and dynamical processes on random graphs.

I. MULTI-TYPE GALTON-WATSON PROCESSES

A. Introduction

The first task concerns the implementation of multi-type Galton-Watson processes and the empirical evaluation of their extinction probabilities.

In particular, the simulation consider $C = 2$ possible classes for the nodes in the tree. An individual v of class c produces a random number of off-springs of class c' , given by the random variable $Y_{c,c'}^{(v)}$, that is distributed according to a **Poisson distribution**:

$$Y_{c,c'}^{(v)} \sim \text{Poisson}(m_{c,c'})$$

Where the mean values $m_{c,c'}$ are defined by a scaling factor α , as it is described in Table I.

TABLE I: Poisson mean values $(m_{c,c'})$ for $\alpha \in \{0.9, 0.95, 1.0, 1.05, 1.1\}$

		Class of the Offspring	
		Class 1	Class 2
Class of the Father	Class 1	$\frac{6}{7}\alpha$	$\frac{2}{7}\alpha$
	Class 2	$\frac{2}{7}\alpha$	$\frac{3}{7}\alpha$

B. Experiments and Results

To evaluate the asymptotic extinction probability q , the following approach is adopted:

- 1) **Parameter Initialization:** Select a scaling factor $\alpha \in \{0.9, 0.95, 1.0, 1.05, 1.1\}$ and construct the mean reproduction matrix $\mathbf{M}$.
- 2) **Simulation Setup:** Execute N_{sim} independent Monte Carlo realizations of the multi-type Galton-Watson process. Each realization is initialized at generation $t = 0$ with a single ancestor of a fixed class c_{init} .
- 3) **Branching Process:** For each generation t up to a maximum of T_{max} , every individual of class i produces offspring of class j independently, according to a Poisson distribution with mean M_{ij} .
- 4) **Extinction Criterion:** A realization is recorded as an *extinction event* if the total population size becomes zero ($|P_t| = 0$) at any step $t \leq T_{max}$.

- 5) **Estimation:** Compute the asymptotic extinction probability $\hat{q}$ as the ratio of the number of extinction events to the total number of simulations:

$$\hat{q} = \frac{E}{N_{sim}} \quad (1)$$

where E is the count of simulations that ended in extinction.

The detailed implementation, for a fixed value of α , is described in Algorithm 1.

Algorithm 1 Monte Carlo Extinction Estimation (α fixed)

Require: Matrix $\mathbf{M}$, T_{max} , N_{sim} , Class c_{init}

Ensure: Est. extinction prob $\hat{q}$

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1:  $E \leftarrow 0$ 
2: for  $n = 1$  to  $N_{sim}$  do
3:    $P \leftarrow$  list with one individual of class  $c_{init}$ 
4:    $is\_extinct \leftarrow \text{False}$ 
5:   for  $t = 1$  to  $T_{max}$  do
6:     if  $P$  is empty then
7:        $is\_extinct \leftarrow \text{True}$ 
8:       break
9:     end if
10:     $P_{next} \leftarrow \emptyset$ 
11:    for each individual  $i \in P$  do
12:       $c \leftarrow$  class of  $i$ 
13:      Draw  $n_1 \sim \text{Poisson}(\mathbf{M}_{c,1})$ 
14:      Draw  $n_2 \sim \text{Poisson}(\mathbf{M}_{c,2})$ 
15:      Add  $n_1$  of Class 1 to  $P_{next}$ 
16:      Add  $n_2$  of Class 2 to  $P_{next}$ 
17:    end for
18:     $P \leftarrow P_{next}$ 
19:  end for
20:  if  $is\_extinct$  then
21:     $E \leftarrow E + 1$ 
22:  end if
23: end for
24:  $\hat{q} \leftarrow E/N_{sim}$ 
25: return  $\hat{q}$ 

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In the experiments, the number of independent runs for each value of α is set to $N_{sim} = 100$, while the maximum number of generations in a single run is set to $T_{max} = 100$.

The results are summarized in Table II and Table III. It can be seen that for both the initial conditions, when $\alpha < 1$ then the extinction is guaranteed, as it is in the **subcritical** case for the single-type Galton-Watson processes when $m < 1$. Instead, extinction is not assured for $\alpha > 1$, and it become less likely as α increases, as in the case of **supercritical** regime.

C. Analysis of the 2-Type Galton-Watson Process

The Galton-Watson process is analyzed here in the general case of 2 types. This process is defined as a Markov

TABLE II: Simulation results for $c_{init} = \text{Class 1}$.

α	Extinction Probability ($\hat{q}$)	Avg. Last Generation ($\bar{T}_{last}$)
0.90	1.00	5.03
0.95	1.00	4.68
1.00	0.98	9.20
1.05	0.89	15.49
1.10	0.75	28.25

TABLE III: Simulation results for $c_{init} = \text{Class 2}$.

α	Extinction Probability ($\hat{q}$)	Avg. Last Generation ($\bar{T}_{last}$)
0.90	1.00	3.68
0.95	1.00	2.84
1.00	1.00	5.03
1.05	0.93	9.63
1.10	0.88	14.99

chain $\{Z_n\}_{n \geq 0}$ on the state space $T \subseteq \mathbb{R}^2$ (vectors with non-negative integer components, representing the number of individuals for each type in a generation), initialized with a single ancestor $Z_0 \in \{e_1, e_2\}$ (standard basis of $\mathbb{R}^2$).

The process, as already seen, evolves through independent reproduction events where an individual of type i (for $i \in \{1, 2\}$) produces a random number of offspring of type 1 and type 2. It is assumed, also in this general case, that the offspring generation is governed by Poissonian processes, where the number of descendants follows a Poisson distribution defined by specific rate parameters.

1) *Mean Matrix and Population Growth*: To characterize the average growth of the population, the mean matrix M is introduced. The entries of this matrix are defined by the expected number of type j offspring produced by a single parent of type i :

$$[M]_{ij} = \mathbb{E}[Z_1^{(j)} \mid Z_0 = e_i] \quad (2)$$

where $Z_1^{(j)}$ denotes the number of type j individuals in generation 1. Due to the independence of reproduction, the expectation of the total population in the next generation is the sum of the expected offspring from the current generation. Relying on the linearity of expectation, the average population vector at generation $n + 1$ is obtained by multiplying the average population at the previous generation by the mean matrix:

$$\mathbb{E}[Z_{n+1}] = \mathbb{E}[Z_n]M \quad (3)$$

By iterating this relationship recursively starting from the initial ancestor Z_0 , the expected population vector at generation n is directly related to the initial state through the n -th power of the mean matrix:

$$\mathbb{E}[Z_n] = Z_0 M^n \quad (4)$$

2) *Probability Generating Functions and Extinction*: While the mean matrix describes the average behavior, the full probability distribution is analyzed using probability generating

functions. Let $\mathbf{f} = (f^{(1)}, f^{(2)})$ [2] be the vector of generating functions for the direct descendants, defined by:

$$f^{(i)}(s_1, s_2) = \sum_{(r_1, r_2) \in T} p^{(i)}(r_1, r_2) s_1^{r_1} s_2^{r_2}, \quad i \in \{1, 2\} \quad (5)$$

where $p^{(i)}(r_1, r_2)$ represents the probability that a type i individual has r_1 descendants of type 1 and r_2 descendants of type 2 according to the underlying Poisson distributions.

The population at the next generation Z_{n+1} is the sum of independent random variables representing the offspring of the individuals in Z_n . This structure implies that the generating function for the n -th generation, denoted $\mathbf{f}_n$, is obtained by the n -fold composition of $\mathbf{f}$ [2]. That is, the generating function for generation $n + 1$ is found by the following equation:

$$\mathbf{f}_{n+1}(\mathbf{s}) = \mathbf{f}(\mathbf{f}_n(\mathbf{s})) \quad (6)$$

The long-term behavior is determined by the extinction probability vector $\mathbf{q} = (q_1, q_2)$, where each component represent the extinction probability starting from a different initial ancestor. The vector $\mathbf{q}$ is identified as the smallest non-negative fixed point of the generating function [2], like in the single-type case:

$$\mathbf{q} = \mathbf{f}(\mathbf{q}) \quad (7)$$

Assuming the moment matrix M is positive regular, the Perron-Frobenius theorem guarantees the existence of a maximal eigenvalue ρ . This eigenvalue determines the criticality of the process [2], analogous to the scalar mean m in the single-type case:

- If $\rho \leq 1$, then $\mathbf{q} = (1, 1)$, indicating that extinction occurs with probability one.
- If $\rho > 1$, then $\mathbf{q} < (1, 1)$ (component-wise), indicating a positive probability of survival.

This theoretical analysis is also confirmed by experimental results. Indeed, the mean matrix shown in Table I is positive regular, and its maximum eigenvalue (ρ) is equal to α . This implies the following conclusions, which have already been verified:

- If $\alpha < 1$, then $\mathbf{q} = (1, 1)$.
- If $\alpha > 1$, then $\mathbf{q} < (1, 1)$.

II. EPIDEMIC PROCESSES

A. Model Design

The objective of the second task is to design and implement a compartmental model that simulates the dynamics of an epidemic.

The proposed model has 7 compartments:

- 1) **Susceptible** (S): people who can be infected.
- 2) **Asymptomatic** (A): people who have been infected, but do not show the symptoms.
- 3) **Symptomatic** (I): infected people who show the symptoms.
- 4) **Quarantined** (Q): people who have been quarantined, or in any case isolated from the rest of the population.

- 5) **Recovered** (R): people who recovered from the infection and are temporarily immune.
- 6) **Dead** (D): people who died because of the infection.
- 7) **Vaccinated** (V): people who have been vaccinated against the infection.

Additionally, the total population is divided in different groups based on 3 factors:

- 1) **Geographic Location**: Turin City, Turin Metropolitan Area, Piedmont.
- 2) **Age**: less than 65 y.o., more than 65 y.o.
- 3) **Exposure Level**: high risk (e.g., doctors), medium risk (e.g., teachers), low risk (e.g., people working from home).

The combination of these factors generates 18 heterogeneous groups, which differ from each other in many ways: the urban population has a higher contact rate than the rural population, the elderly are subject to a higher mortality rate, and doctors are more exposed to infection than housewives.

The transitions between the different compartments are regulated by various hyperparameters. Firstly, a transmission rate table ($\mathbf{B} \in \mathbb{R}^{18 \times 18}$) is calculated as:

$$[\mathbf{B}]_{ij} = \beta_{ij} \quad \text{rate at which group } j \text{ infects group } i$$

A susceptible person in group k is subject to a total force of infection λ_k that sum contributions from all other groups:

$$\lambda_k = \sum_{j=1}^{18} \beta_{kj} \cdot \frac{I_j + \epsilon A_j}{N_j} \quad (8)$$

where $\epsilon < 1$ is a scaling factor for asymptomatic infectivity and N_j represents the living population in group j .

When a susceptible person gets infected he transitions to the asymptomatic compartment (A). Then, after a certain latency σ , the infection becomes symptomatic and he transitions to the correspondent compartment (I). If screening testing is conducted by public-health officers, then an asymptomatic individual can also be isolated, i.e., transition to quarantined (with rate θ_A).

An infected and symptomatic person can either die (with a rate $\delta_I^{(k)}$ dependent on his demographic group k), recover from the infection (with a recovery rate γ_I) or be quarantined (with rate $\theta_I > \theta_A$), if some public-health policies are applied.

Quarantined people can also recover from their infection, with a rate $\gamma_Q > \gamma_I$, since it is supposed that some medical interventions will be performed on the isolated patients to improve their condition faster. Otherwise, they can die, with a demographic dependent rate $\delta_Q^{(k)} < \delta_I^{(k)}$, for the same reason as before.

A person that recovered from the infection gains a temporary immunity, then he will be once again susceptible, with a transition rate ξ .

Finally, if a vaccination campaign is in progress, susceptible people will be vaccinated with a fixed maximum rate v , transitioning to the vaccinated compartment. As for the recovery,

also the vaccination has a limited effect in time, so vaccinated people will transition back to the susceptible compartment, with a rate $\xi_V < \xi$, since it is supposed that the vaccination provides a longer immunity than the natural recovery.

Figure 1 summarizes all possible transitions between compartments.

B. Experiments and Results

For the first part of the analysis, no vaccination campaign will be considered. The **mean field equations** that regulate the simulation are the following:

$$\frac{dS_k}{dt} = -S_k \lambda_k + \xi R_k \quad (9)$$

$$\frac{dA_k}{dt} = S_k \lambda_k - (\sigma + \theta_A) A_k \quad (10)$$

$$\frac{dI_k}{dt} = \sigma A_k - (\theta_I + \gamma_I + \delta_I^{(k)}) I_k \quad (11)$$

$$\frac{dQ_k}{dt} = \theta_I I_k + \theta_A A_k - (\gamma_Q + \delta_Q^{(k)}) Q_k \quad (12)$$

$$\frac{dR_k}{dt} = \gamma_I I_k + \gamma_Q Q_k - \xi R_k \quad (13)$$

$$\frac{dD_k}{dt} = \delta_I^{(k)} I_k + \delta_Q^{(k)} Q_k \quad (14)$$

The **baseline** simulation to be used as a reference will not take into account any public health policies, i.e., **neither quarantine** ($Q = 0$) **nor lockdown**. The hyperparameters, optimized to be both realistic and meaningful for the simulation, are summarized in Table IV. It is assumed that the epidemic starts with 10 infected people only among the group of individuals residing in the city, with less than 65 y.o. and with high exposure.

Figure 2 and Table V show the results for this baseline execution. It is clear that without any intervention the effects of the epidemic are quite devastating for the population, reaching a mortality rate of over 6%.

In the following simulations some public-health policies are applied. Figure 3 and Table VI show the effect of an **early lockdown** (from day 20 to day 120). During the lockdown period, the transmission rate for the infection is reduced by 70% ($\mathbf{B}_{lockdown} = 0.3 \cdot \mathbf{B}$). It can be observed that the lockdown contributes to reducing the peak of infection and also mortality, but only slightly.

Figure 4 and Table VII, on the other hand, show the effect of isolating infected individuals (i.e., **quarantine**) with rate $\theta_I = 0.15$ for people who show symptoms and a lower rate $\theta_A = 0.005$ for asymptomatic people. In this scenario the improvement with respect to the baseline is apparent: the peak of symptomatic is reduced by more than 50% and the mortality rate is reduced by 60%, also the attack rate (percentage of the total population getting infected at some point) is drastically decreased.

Finally, Figure 5 and Table VIII show the combined effect of **lockdown and quarantine**. These results are very close to those obtained by applying quarantine alone, confirming

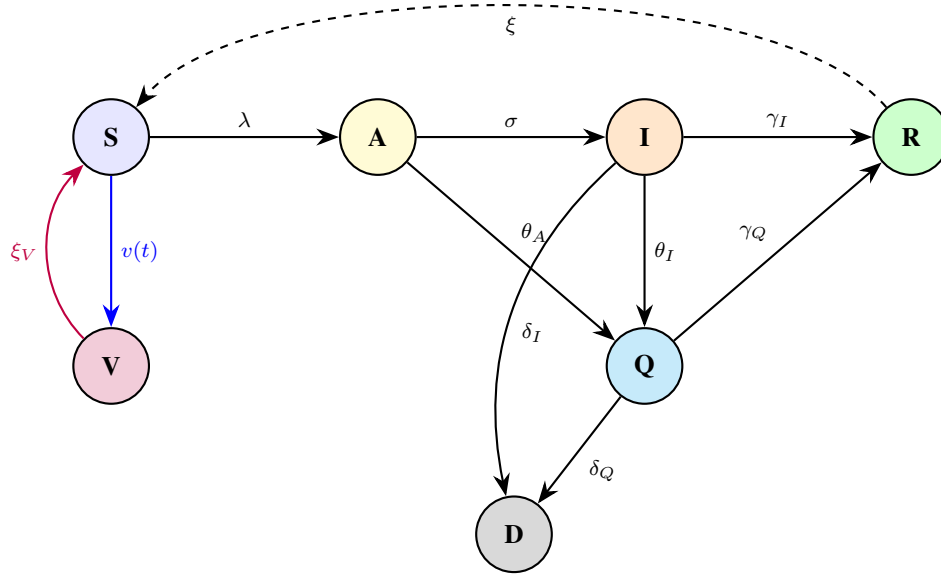


Fig. 1: **Schematic of the SAIQRS-V Epidemic Model.** Nodes: S: Susceptible, A: Asymptomatic, I: Symptomatic/Infected, Q: Quarantined, R: Recovered, D: Dead, V: Vaccinated.

TABLE IV: Simulation Hyperparameters - Baseline Execution

Parameter	Value	Description
<i>Biological & Clinical Dynamics</i>		
σ (Sigma)	$1/5.0 \text{ day}^{-1}$	Latency rate (progression $A \rightarrow I$)
ϵ (Epsilon)	0.4	Relative infectivity of Asymptomatic nodes
ξ (Xi)	$1/120.0 \text{ day}^{-1}$	Natural immunity waning rate ($R \rightarrow S$)
γ_I	$1/10.0 \text{ day}^{-1}$	Recovery rate from Symptomatic state
γ_Q	$1/8.0 \text{ day}^{-1}$	Recovery rate from Quarantined state
θ_I	0.0	Quarantine rate for symptomatic ($I \rightarrow Q$)
θ_A	0.0	Quarantine rate for asymptomatic ($A \rightarrow Q$)
δ_I (Young/Old)	0.0002 / 0.005	Fatality rate for Symptomatic individuals
δ_Q (Young/Old)	0.00005 / 0.001	Fatality rate for Quarantined individuals
<i>Transmission & Contact Structure</i>		
Base Transmission	0.50, 0.05, 0.02	Baseline β for City, Metro, and Piedmont
f_{geo} (Mobility)	0.20 / 0.05	Contact multiplier for City-Metro / Other regions
f_{age} (Mixing)	1.2 / 0.8 / 0.6	Contact multiplier for Young-Young / Old-Old / Mixed
f_{work} (Risk Map)	2.5, 0.8, 0.2	Exposure factors for High, Mid, and Low exp.
<i>Simulation Settings</i>		
T_{max}	1095 days	Total duration of the simulation

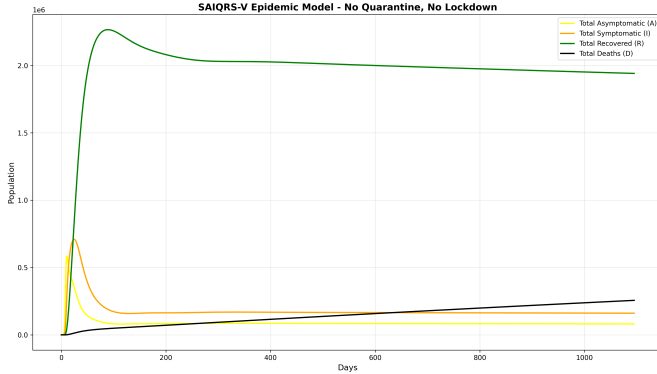


Fig. 2: Trajectory of the overall population for the baseline simulation.

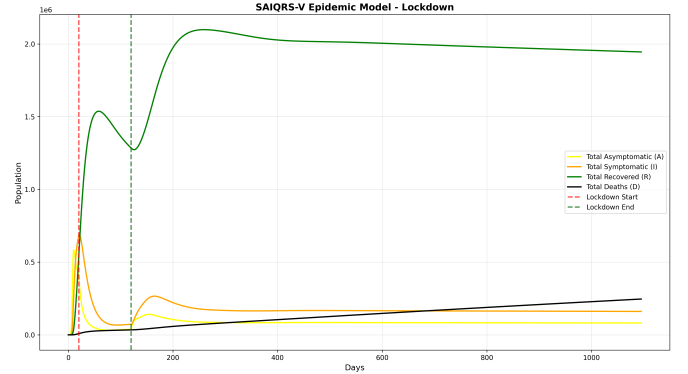


Fig. 3: Trajectory of the overall population for the early lockdown simulation.

TABLE V: Summary of Simulation Results (Baseline)

Metric	Value
Total Population	4,200,000
Peak Symptomatic	709,434 (Day 24)
Peak Asymptomatic	583,257 (Day 10)
Total Deaths	256,163
Final Recovered	1,941,429
Attack Rate	52.32%
Deaths per 100k	6,099.1

TABLE VI: Summary of Simulation Results (Early Lockdown)

Metric	Value
Total Population	4,200,000
Peak Symptomatic	689,039 (Day 21)
Peak Asymptomatic	583,257 (Day 10)
Total Deaths	246,275
Final Recovered	1,945,293
Attack Rate	52.18%
Deaths per 100k	5,863.7

once again that isolating infected people is one of the most significant measures to reduce the risk of an epidemic.

C. Vaccination Strategies

For this second part of the analysis, it is assumed that a **vaccine** becomes available after a certain number of days from the start of the epidemic. In particular the hyperparameters related to the vaccine are summarized in Table IX.

Since quarantine has proven to be the most effective non-pharmacological intervention, infected people will also be partially isolated in these simulations.

With the introduction of a **Vaccinated** compartment, the new **mean field equations** are the following:

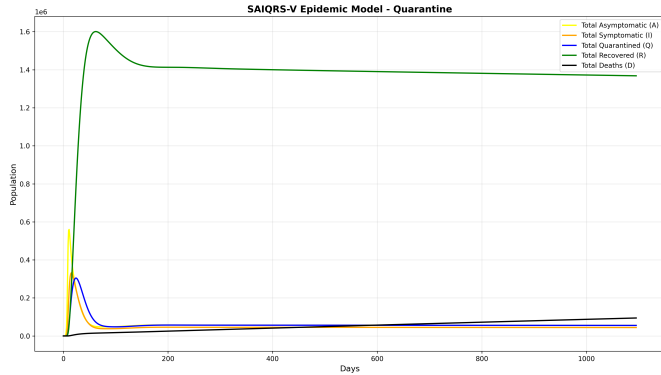


Fig. 4: Trajectory of the overall population for the quarantine simulation.

TABLE VII: Summary of Simulation Results (Quarantine)

Metric	Value
Total Population	4,200,000
Peak Symptomatic	331,758 (Day 15)
Peak Asymptomatic	560,088 (Day 11)
Total Deaths	94,499
Final Recovered	1,368,211
Attack Rate	34.83%
Deaths per 100k	2,250.0

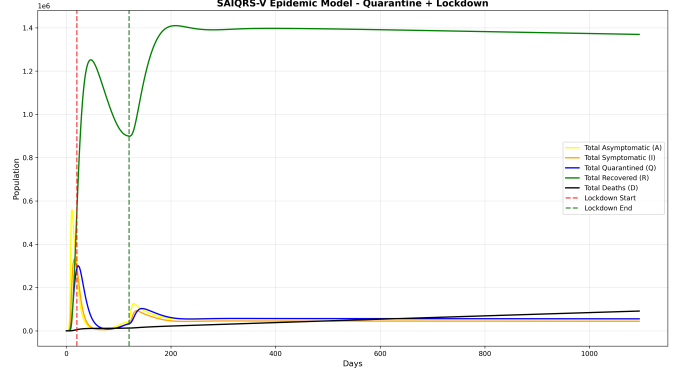


Fig. 5: Trajectory of the overall population for the quarantine with lockdown simulation.

$$\frac{dS_k}{dt} = -S_k \lambda_k + \xi R_k + \xi_V V_k - v_k(t) \quad (15)$$

$$\frac{dA_k}{dt} = S_k \lambda_k - (\sigma + \theta_A) A_k \quad (16)$$

$$\frac{dI_k}{dt} = \sigma A_k - (\theta_I + \gamma_I + \delta_I^{(k)}) I_k \quad (17)$$

$$\frac{dQ_k}{dt} = \theta_I I_k + \theta_A A_k - (\gamma_Q + \delta_Q^{(k)}) Q_k \quad (18)$$

$$\frac{dR_k}{dt} = \gamma_I I_k + \gamma_Q Q_k - \xi R_k \quad (19)$$

$$\frac{dD_k}{dt} = \delta_I^{(k)} I_k + \delta_Q^{(k)} Q_k \quad (20)$$

$$\frac{dV_k}{dt} = v_k(t) - \xi_V V_k \quad (21)$$

TABLE VIII: Summary of Simulation Results (Quarantine + Lockdown)

Metric	Value
Total Population	4,200,000
Peak Symptomatic	331,758 (Day 15)
Peak Asymptomatic	560,088 (Day 11)
Total Deaths	91,394
Final Recovered	1,369,389
Attack Rate	34.78%
Deaths per 100k	2,176.0

TABLE IX: Vaccination Hyperparameters

Parameter	Value	Description
T_{start_vax}	365	Day on which the vaccination campaign begins
v_{total}	4000 doses/day	Total daily vaccination capacity available for the entire population
ξ_V	$1/365.0 \text{ day}^{-1}$	Rate of vaccine immunity waning (approx. 1 year duration)

Several vaccination strategies are implemented, namely:

- **Uniform:** all demographic groups receive a number of vaccines proportional to their size.
- **Old First:** vaccination is a priority for the elderly population, which means that only after all susceptible individuals over the age of 65 have been vaccinated will it be possible to vaccinate younger people as well.
- **High Exposure First:** the vaccination campaign begins with the highest exposure groups, then moves on to those at medium risk, and finally to those at low risk of exposure.
- **Geographical:** vaccines are being provided first to the urban population, then to the rest of the metropolitan population, and finally to the rest of the region.

Figure 6 and Table X show the comparative results for each vaccination strategy. The first important conclusion is that any vaccination campaign is significantly effective in reducing the negative effects of the epidemic, since even in the worst-case scenario (i.e., uniform vaccination), the total number of deaths is reduced by more than 10%.

Figure 7 focuses on the analysis of total deaths, which represent the most critical factor in an epidemic. It can be observed that the system's behavior is obviously the same before the start of the vaccination campaign, but the mortality trajectories diverge subsequently, demonstrating that uniform vaccination is less effective, while the **Geographic strategy** is the most effective in reducing epidemic damage. Specifically, applying a geographic vaccination strategy reduces the total number of deaths by 46% compared to no vaccination, and the attack rate is reduced by 86%.

To fully understand the mechanism behind the success or failure of each strategy, it is necessary to analyze the demographic dynamics provided in the detailed dashboards. By observing the vaccination progress (Figures 8, 9, and 10), the distinct allocation priorities of each strategy can be correlated with their epidemiological outcomes shown in the comparison dashboards (Figures 11, 12, 13, and 14).

The **Old First** strategy prioritizes the direct protection of the most vulnerable. As observed in Figure 8, the vaccination curve for the “Old” group rises sharply at day 365, while the “Young” group remains unvaccinated until the elderly are fully covered. The consequence is evident in Figure 11 (Age Group Comparison): the cumulative death rate for the Old population flattens almost instantly compared to the No Vaccination baseline. However, the symptomatic infection rate among the Young (who are the primary drivers of mobility) continues to track closely with the baseline. This explains why this strategy minimizes mortality effectively but fails to significantly reduce the overall attack rate (it remains greater than 20%).

Conversely, the **High Exposure First** strategy targets the active workforce. Figure 10 shows the vaccination coverage saturating the “High Exposure” groups first. The impact of this approach is best analyzed in Figure 13 (Exposure Level Comparison), where it can be seen that the infection curves

for high-exposure individuals are suppressed early. By neutralizing these “super-spreaders,” the strategy reduces the viral transmission momentum for the entire population, resulting in a significantly lower overall Attack Rate (14.9%) compared to the Old First strategy (20.9%). However, a trade-off is observed in Figure 11: since the elderly population must wait for the high-risk workers to be immunized, their cumulative death rate remains higher than in the Old First scenario.

The **Geographical** strategy demonstrates the power of network-based intervention. Figure 9 indicates that the vaccination effort is concentrated initially on the City population. The epidemiological effect is illustrated in Figure 12 (Geographical Comparison): by immunizing the City, which acts as the central hub of the epidemic network, the infection curves for the Metro and Piedmont regions are also suppressed, despite them receiving vaccines later. This “firebreak” effect is confirmed by the Attack Rate Heatmap (Figure 14), which shows a uniformly low final attack rate across all demographic combinations, achieving the best overall suppression (4.89% Attack Rate).

Finally, the **Uniform** strategy, as shown in the progress dashboards, distributes resources in parallel across all groups. While equitable, Figure 11, Figure 12 and Figure 13 reveal that this approach fails to achieve a tactical advantage in any specific domain. It does not protect the elderly quickly enough to prevent significant mortality, nor does it suppress the transmission hubs in the City or High Exposure groups effectively enough to stop the spread. Consequently, its trajectories in all comparison graphs track closest to the No Vaccination baseline.

The application of a lockdown, aimed at mitigating the spread of the epidemic, during the vaccination campaign has been studied, but this analysis has not produced significant results. During the lockdown period, deaths stabilize, only to increase again as soon as the restrictions are lifted. The final result differs little from that obtained without simulating a lockdown. The results differ more significantly if the restrictions are applied for very long periods, but this is unrealistic, as the effects on society of a lockdown lasting more than six months would still be harmful from other perspectives (e.g., social, economic).

III. DYNAMIC PROCESSES ON RANDOM GRAPHS

A. $G(n,p)$ Graphs

The third and final task involves the creation of different types of random graphs.

The first type of graph to generate is the standard graph $G(n,p)$, consisting of n nodes and in which each possible edge is added, independently, with probability p .

The values of n and p tested are summarized in Table XI. Since a large number of vertices is considered, a more efficient (i.e. linear) algorithm than the naive one is adopted to generate the random graph, exploiting the fact that, assuming that all possible edges (combinations of 2 nodes) are in line, the distance between 2 edges added to the graph is distributed according to a **geometric distribution** with parameter p .

SAIQRS-V Epidemic Model - Vaccination Strategy Comparison

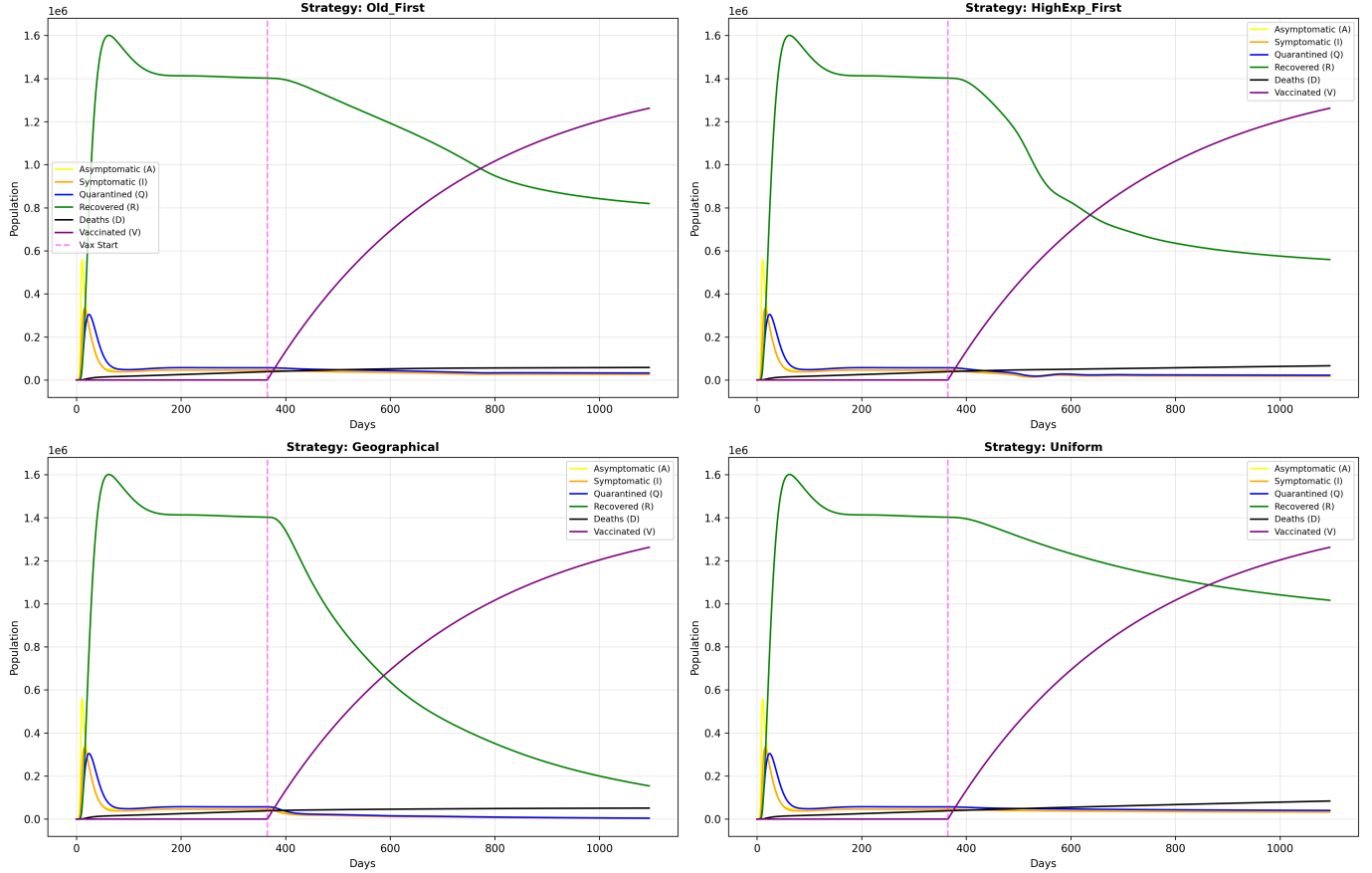


Fig. 6: Comparative results for the different types of vaccination strategies.

TABLE X: Comparative Analysis of Vaccination Strategies

Strategy	Total Deaths	Deaths per 100k	Attack Rate (%)	Deaths Reduced (%)	Attack Rate Reduced (%)	Final Immune (%)
<i>No Vaccination</i>	94,499	2,250.0	34.83	—	—	32.58
Old First	58,175	1,385.1	20.90	38.4%	40.0%	49.57
High Exp. First	66,105	1,573.9	14.90	30.0%	57.2%	43.38
Geographical	51,040	1,215.2	4.89	46.0%	86.0%	33.73
Uniform	83,946	1,998.7	26.20	11.2%	24.8%	54.26

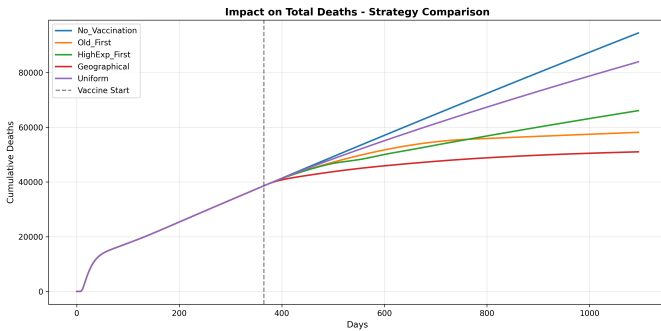


Fig. 7: Comparative results of the total number of deaths with different vaccination strategies.

The detailed implementation of the generation is described in Algorithm 2.

TABLE XI: Simulation Parameters - Linear $G(n,p)$

Parameter	Description / Values
Population Size (n)	$\{10^4, 10^5, 10^6\}$
Probability Formula	$p = a/n$
Coefficient (a)	0.9, 0.95, 0.98, 1.0, 1.02, 1.05, 1.1, 1.2

The results of this first batch of simulations are summarized in Table XII, for each combination of n and a 10 independent runs are executed (the results are averaged). The simulation results confirm the phase transition properties predicted by the Galton-Watson branching process theory [1] ($p = a/n \iff \mu = a$).

- **Subcritical Regime** ($a < 1$): The process is extinct with

Vaccination Progress by Age Group Across Strategies

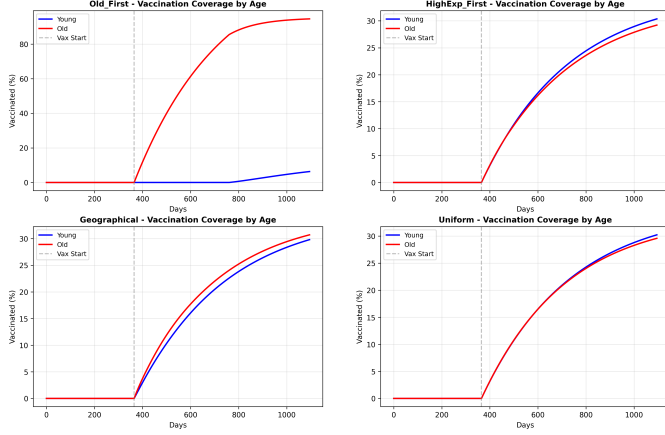


Fig. 8: Temporal evolution of vaccination coverage differentiated by age group (Young vs. Old) for each strategy.

Vaccination Progress by Exposure Risk Level Across Strategies

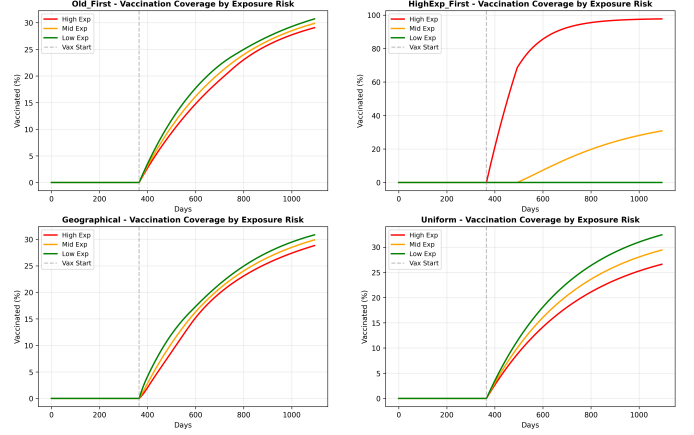


Fig. 10: Temporal evolution of vaccination coverage differentiated by exposure level (High, Mid, Low) for each strategy.

Vaccination Progress by Geographic Region Across Strategies

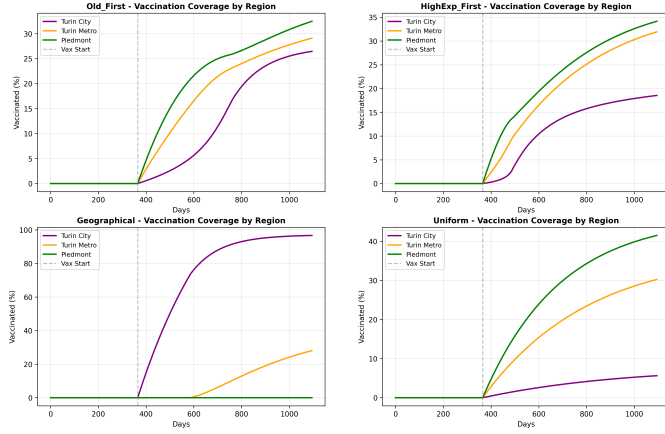


Fig. 9: Temporal evolution of vaccination coverage differentiated by geographical location (City, Metro, Piedmont) for each strategy.

probability 1. The largest component scales logarithmically, $|C_1| \propto K_a \log n$.

- *Note on Dimensions:* While values like $|C_1| \approx 474$ (for $a = 0.9, n = 10^6$) may seem large compared to $\ln(10^6) \approx 13.8$, the *scaling* is indeed logarithmic. As n increases by factors of 10 ($10^4 \rightarrow 10^5 \rightarrow 10^6$), the component size increases by roughly constant additive steps ($\approx +170$), not multiplicative factors. The large absolute values result from the pre-factor K_a being large due to $a = 0.9$ being close to the critical threshold.

- **Critical Regime ($a = 1$):** The system follows a power law, and the largest component scales as $|C_1| \propto n^{2/3}$. For $n = 10^6$, the observed $|C_1| \approx 10,237$ matches the theoretical prediction (10^4) almost perfectly.
- **Supercritical Regime ($a > 1$):** A unique *giant compo-*

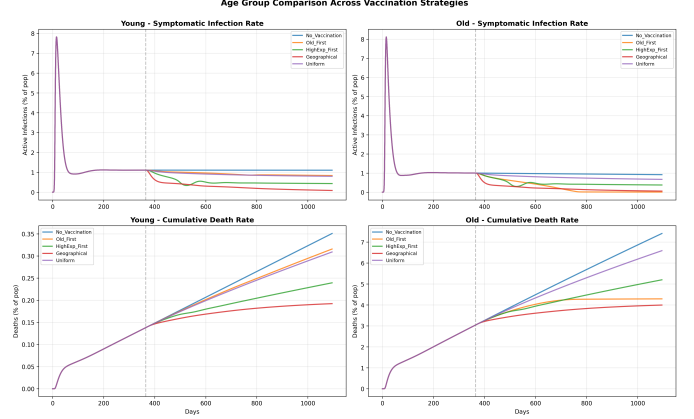


Fig. 11: Comparative analysis of symptomatic infection rates (left) and cumulative death rates (right) distinguished by age group.

nent emerges, scaling linearly with system size ($|C_1| \propto n$). At $a = 1.20$ ($n = 10^6$), $|C_1|$ explodes to $\approx 313,000$, while the second largest component $|C_2|$ drops to negligible sizes (≈ 200), confirming the absorption of finite clusters into the giant core.

The same code is now used to generate $G(n,p)$ graphs where p is logarithmically linked to n , namely $p = \frac{a \log(n)}{n}$; the parameters tested are summarized in Table XIII.

The results for this second batch of simulations are reported in Table XIV, also in this case 10 independent runs per configuration have been performed.

In this regime where $p(n) = \frac{a \ln n}{n}$, the graph exhibits a distinct phase transition related to **connectivity** [1], rather than the emergence of a giant component (which is already present for $a > 1/\ln n$).

- **Connectivity Threshold ($a = 1$):** The simulation results confirm the sharp threshold for connectivity at $a = 1$.
 - For $a < 1$ (e.g., $a = 0.90$), the graph is almost

Algorithm 2 Efficient Generation of $G(n, p)$

Require: Number of nodes n , Probability p

Ensure: Graph $G = (V, E)$

```

1:  $V \leftarrow \{0, \dots, n-1\}$ 
2:  $E \leftarrow \emptyset$ 
3: if  $p \leq 0$  then
4:   return  $(V, E)$ 
5: end if
6: if  $p \geq 1$  then
7:   return Complete Graph  $K_n$ 
8: end if
9:  $v \leftarrow 1$ 
10:  $w \leftarrow -1$ 
11: while  $v < n$  do
12:   Draw  $r \sim \text{Uniform}(0, 1)$ 
13:    $s \leftarrow \lceil \ln(r) / \ln(1-p) \rceil$ 
14:    $w \leftarrow w + s$ 
15:   while  $w \geq v$  and  $v < n$  do
16:      $w \leftarrow w - v$ 
17:      $v \leftarrow v + 1$ 
18:   end while
19:   if  $v < n$  then
20:      $E \leftarrow E \cup \{(v, w)\}$ 
21:   end if
22: end while
23: return  $(V, E)$ 

```

TABLE XII: Simulation Results for $G(n, p = a/n)$. Comparison of largest ($|C_1|$) and second largest ($|C_2|$) components across different regimes.

Size (n)	Param (a)	$ C_1 $	$ C_2 $
10^4	0.90	123.8	82.6
	0.95	246.7	124.1
	0.98	218.4	137.9
	1.00	376.9	195.2
	1.02	651.4	224.5
	1.05	1,022.8	211.8
	1.10	1,523.0	227.9
	1.20	3,045.8	74.8
10^5	0.90	285.8	205.4
	0.95	680.3	423.9
	0.98	941.3	617.4
	1.00	2,263.8	764.5
	1.02	2,994.0	1,088.6
	1.05	7,804.9	800.0
	1.10	16,888.1	365.2
	1.20	31,753.3	134.8
10^6	0.90	474.0	394.5
	0.95	1,055.8	804.0
	0.98	2,533.4	1,919.8
	1.00	10,237.1	3,376.9
	1.02	40,040.7	2,348.6
	1.05	92,585.1	1,074.8
	1.10	172,873.0	624.6
	1.20	313,839.0	200.5

TABLE XIII: Simulation Parameters - Logarithmic $G(n, p)$

Parameter	Description / Values
Population Size (n)	$\{10^4, 10^5, 10^6\}$
Probability Formula	$p = \frac{a \ln n}{n}$
Coefficient (a)	0.9, 0.95, 0.98, 1.0, 1.02, 1.05, 1.1, 1.2

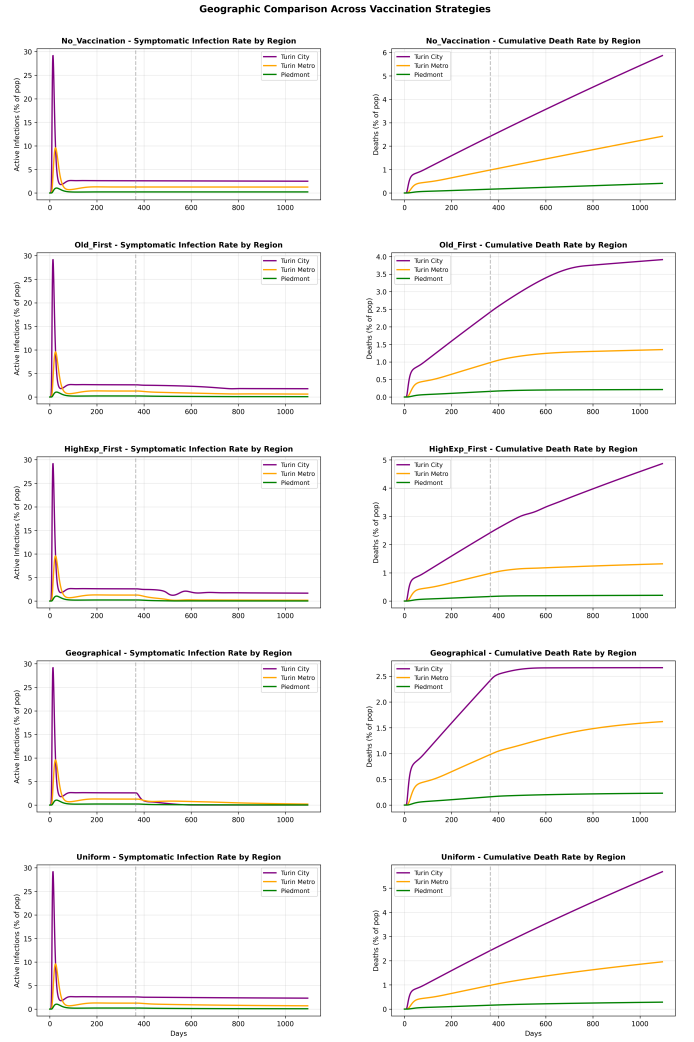


Fig. 12: Comparative analysis of symptomatic infection rates (left) and cumulative death rates (right) distinguished by geographical location.

surely disconnected as $n \rightarrow \infty$. This is evident in the $n = 10^6$ case, where connectivity drops to 0%. The graph contains isolated vertices, preventing full connection.

- For $a > 1$ (e.g., $a = 1.10, 1.20$), the probability of the graph being connected approaches 1. In fact at $n = 10^6$, we observe 100% connectivity for $a = 1.10$.

- **Diameter Evaluation:** When the graph is connected, its diameter is observed to be small, typically scaling as $O(\ln n / \ln(np))$.

- For $n = 10^6$ and $a \geq 1$, the measured diameter is stable around 8. This is extremely compact relative to the population size, highlighting the “small-world” property of random graphs in the connected regime.

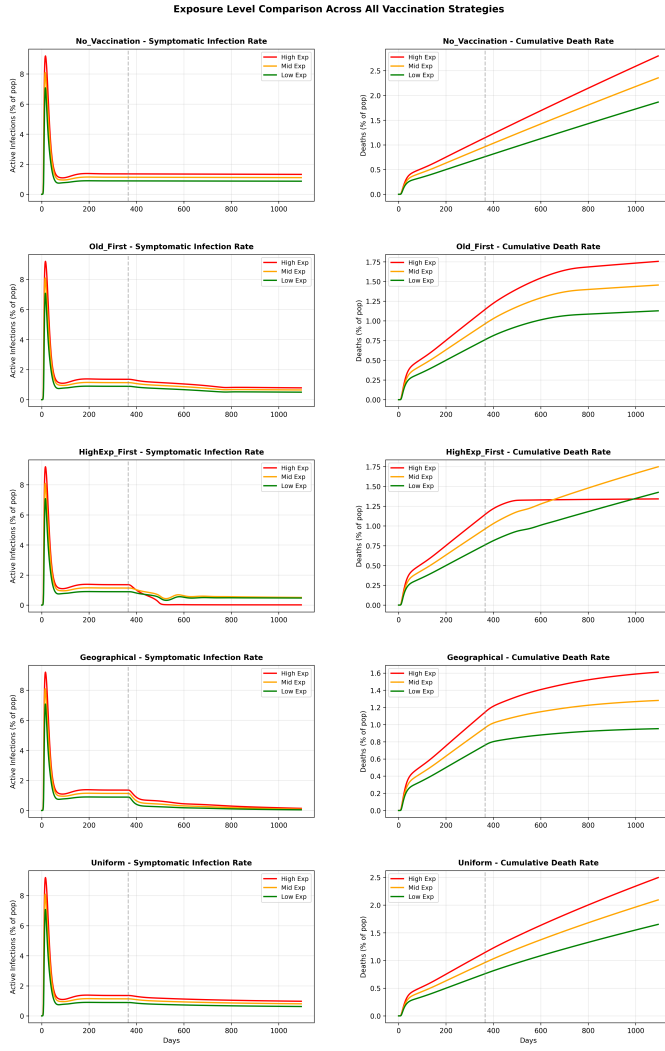


Fig. 13: Comparative analysis of symptomatic infection rates (left) and cumulative death rates (right) distinguished by exposure level.

B. Random Geometric Graphs

The next type of graphs to simulate are the **Random Geometric Graphs** (a.k.a., $G(n,r)$). These graphs have 2 parameters:

- n : number of nodes in the graph.
- r : neighborhood radius; if $\text{distance}(v, w) < r$ then the edge (v, w) is added, for each pair of vertices in the graph.

The values of n and r tested are summarized in Table XV, for each combination of parameters 10 independent runs are executed.

The results for the Random Geometric Graph, shown in Table XVI, highlight two fundamental differences compared to the $G(n,p)$ model, driven by the underlying Euclidean geometry.

- **Connectivity Threshold ($a = 1$):** The simulation results support the theoretical threshold at $a = 1$, but with a

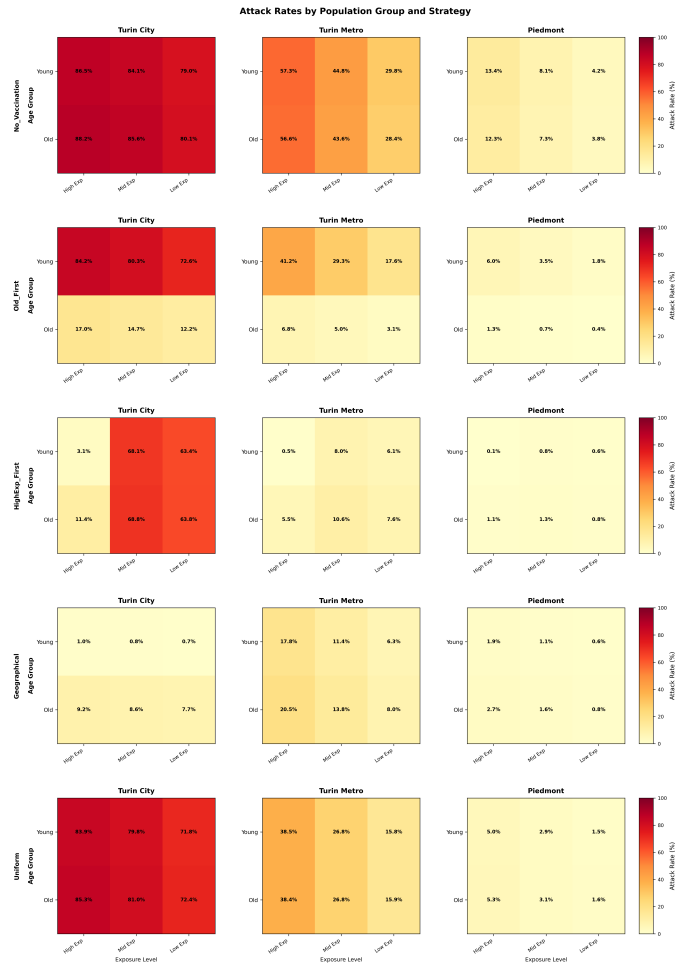


Fig. 14: Heatmap summarizing the final attack rate across all demographic combinations (Geography $\times$ Age $\times$ Exposure) for each vaccination strategy.

TABLE XIV: Simulation Results for $G(n, p = a \ln n/n)$. Connectivity and Diameter.

Size (n)	Param (a)	Connected (%)	Avg. Diameter	$ C_1 $
10^4	0.90	20%	8.00	9,998.4
	0.95	10%	8.00	9,998.3
	0.98	40%	7.00	9,999.1
	1.00	30%	7.00	9,998.9
	1.02	40%	7.50	9,999.2
	1.05	50%	7.20	9,999.5
	1.10	90%	6.67	9,999.8
10^5	1.20	90%	6.22	9,999.9
	0.90	0%	—	99,996.4
	0.95	10%	8.00	99,998.4
	0.98	30%	7.67	99,998.4
	1.00	20%	7.50	99,998.9
	1.02	50%	7.40	99,999.2
	1.05	40%	7.50	99,998.9
10^6	1.10	90%	7.00	99,999.8
	1.20	100%	6.90	100,000.0
	0.90	0%	—	999,995.7
	0.95	10%	9.00	999,998.2
	0.98	0%	—	999,998.3
	1.00	40%	8.00	999,999.1
	1.02	60%	8.00	999,999.4
10^7	1.05	70%	7.86	999,999.7
	1.10	100%	7.80	1,000,000.0
	1.20	90%	7.44	999,999.9

TABLE XV: Simulation Parameters (Geometric Random Graph)

Parameter	Description / Values
Population Size (n)	$\{10^3, 10^4, 10^5, 10^6\}$
Radius Formula (r)	$r = \sqrt{\frac{a \ln n}{\pi n}}$
Coefficient (a)	0.8, 0.9, 0.95, 1.0, 1.05, 1.1, 1.2

notable delay in full connectivity. For $a \leq 1$, the graph is almost never connected, despite the giant component $|C_1|$ encompassing $> 99\%$ of the nodes. This behavior is due to geometric constraints at the domain boundaries: nodes located near the edges (and especially in the corners) of the unit square have a reduced effective interaction area, as part of their radius falls outside the domain. Consequently, they have lower expected degrees and are statistically more likely to remain isolated than nodes in the bulk, preventing full connectivity until a is sufficiently large.

- **Macroscopic Diameter** ($O(\sqrt{n/\ln n})$): The most striking difference is the diameter. In the $G(n, p)$ model (logarithmic regime), the diameter for $n = 10^6$ was extremely small (≈ 8), whereas in the RGG model, it is massive (≈ 727).

This large diameter can be formally justified by a simple geometric argument. In an RGG on the unit square, an edge exists only if the Euclidean distance satisfies $d(u, v) < r$. The maximum physical distance between two nodes is the diagonal length $\sqrt{2}$. To traverse this distance, a path must consist of k hops, where each hop covers at most distance r . Therefore, the minimum diameter D_{min} is bounded by $D_{min} \cdot r > \sqrt{2}$. Substituting the radius formula $r = \sqrt{\frac{a \ln n}{\pi n}}$, we obtain:

$$D \sim \frac{\sqrt{2}}{\sqrt{\frac{a \ln n}{\pi n}}} = \sqrt{\frac{2\pi}{a}} \cdot \sqrt{\frac{n}{\ln n}}$$

This derivation confirms that the diameter scales as $O(\sqrt{n/\ln n})$. For $n = 10^6$, this scaling predicts a value in the hundreds, aligning perfectly with the simulation result of 727.

C. Majority Model on RGG

The final part of this laboratory involves simulating a majority model on a random geometric graph with parameters $n = 10^4$ and $r = \sqrt{\frac{3 \ln n}{\pi n}}$. At time $t = 0$, nodes are initialized with an opinion of +1 with probability $p \in 0.5, 0.6, 0.7, 0.8$. When a node v wakes-up, that node's opinion changes based on the majority opinion of its neighbors (ties are broken randomly). Each node is associated with an opinion that can be +1 or -1, and the wake-up times for each node are obtained from a Poissonian process with parameter $\lambda_v = 1$.

The results are shown in Figure 15 and Table XVII, for each value of p , 20 independent runs are executed for a maximum time $T_{max} = 100$.

By analyzing the results it appears to be a phase transition between $p = 0.6$ and $p = 0.7$.

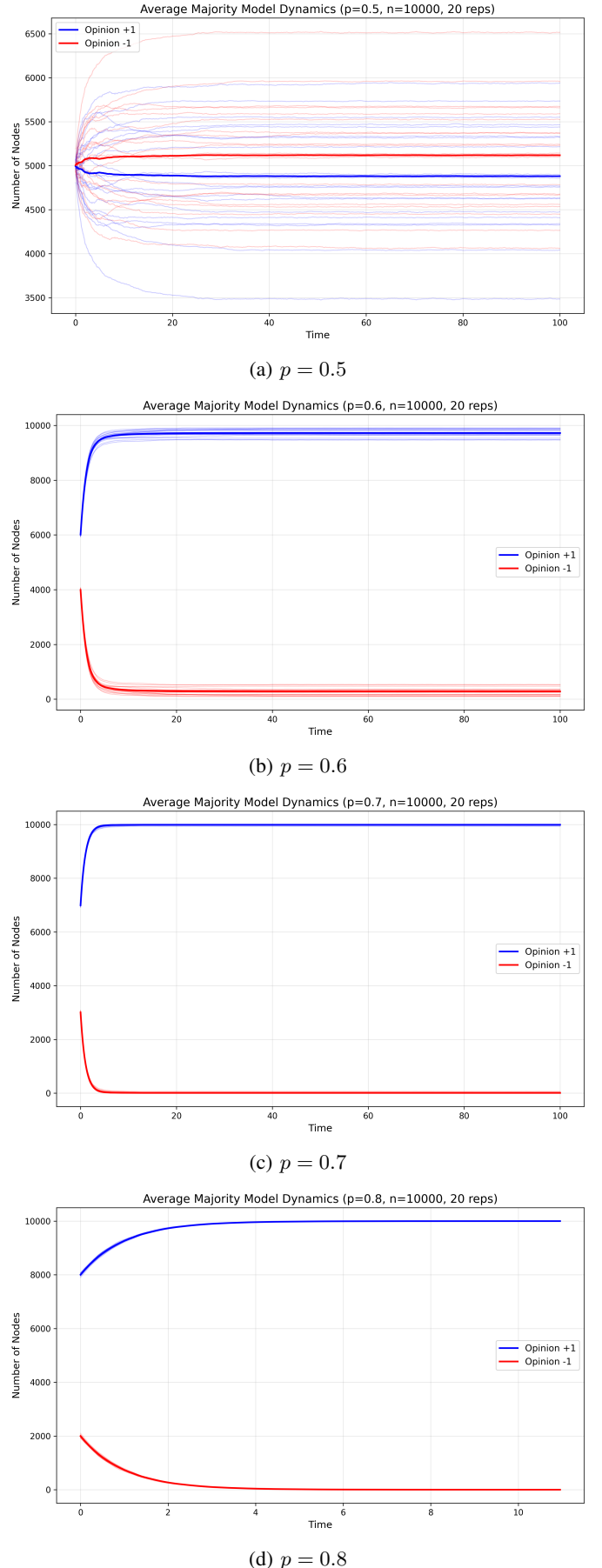


Fig. 15: Trajectories of the average number of nodes for each opinion across varying probabilities p .

TABLE XVI: Simulation Results for RGG with $r = \sqrt{\frac{a \ln n}{\pi n}}$.

Size (n)	Param (a)	Connected (Count)	Avg. Diameter	$ C_1 $
10^3	0.80	0	—	836.2
	0.90	0	—	967.7
	0.95	0	—	976.0
	1.00	0	—	981.7
	1.05	0	—	988.5
	1.10	1	38.0	991.6
	1.20	2	35.5	997.0
10^4	0.80	0	—	9,960.7
	0.90	0	—	9,986.3
	0.95	0	—	9,987.7
	1.00	0	—	9,991.0
	1.05	0	—	9,993.8
	1.10	1	101.0	9,996.3
	1.20	6	95.7	9,999.1
10^5	0.80	0	—	99,960.3
	0.90	0	—	99,987.3
	0.95	0	—	99,994.2
	1.00	0	—	99,993.7
	1.05	2	284.5	99,996.1
	1.10	1	274.0	99,998.2
	1.20	4	257.8	99,999.0
10^6	0.80	0	—	999,955.3
	0.90	0	—	999,989.0
	0.95	1	850.0	999,994.0
	1.00	0	—	999,995.0
	1.05	2	791.5	999,998.4
	1.10	5	767.8	999,999.0
	1.20	6	727.3	999,999.5

TABLE XVII: Majority Model Dynamics on RGG ($n = 10^4, r \approx 0.03$).

Initial Prob. (p)	Consensus Rate	Avg. Time
0.5	0.00	—
0.6	0.00	—
0.7	0.80	8.99
0.8	1.00	8.35

- **Metastability at $p \leq 0.6$:** For these values, the system fails to reach consensus within the simulation time frame, which is actually very large. Since all simulated graphs are fully connected, this behavior cannot be explained by different connected components reaching different consensus opinions.

Therefore, the reason for these results must lie in the distribution of initial opinions; evidently, some **clusters** of nodes with opinion +1 and other clusters with opinion -1 form, preventing a global consensus from being reached. This hypothesis is further demonstrated by the fact that, as can be seen in Figure 15b, with $p = 0.6$ a stable configuration is reached, in which the vast majority of nodes have opinion +1, but some very small clusters still have their opinion fixed at -1.

- **Rapid Convergence at $p \geq 0.7$:** Above this threshold, the system begins to **converge** reliably. Clusters of opinions equal to -1 become very unlikely to form, so much so that, for $p = 0.8$, consensus is reached in 100% of the trials.

Once the initial base is strong enough to trigger a global consensus, convergence is relatively rapid (≈ 8), decreasing slightly as the initial majority strengthens.

REFERENCES

- [1] Paul Erdős, Alfréd Rényi, et al. On the evolution of random graphs. *Publications of the*, 1960.
- [2] ADITYA SINGH. The galton-watson branching process. 2024.